

第二次

1. 证明:

对任意 $x_1, x_2 \in S$, $\exists y_1, y_2 = 0$

$$\begin{cases} x_1 = Ay_1 \\ x_2 = Ay_2 \end{cases} \Rightarrow \lambda x_1 + (1-\lambda)x_2 = \lambda Ay_1 + A(1-\lambda)y_2 = A(\lambda y_1 + (1-\lambda)y_2) = A y_3$$

$\lambda y_1 + (1-\lambda)y_2 \geq 0 \Rightarrow \lambda x_1 + (1-\lambda)x_2 \in S$

$\therefore S$ 为凸集

2. 证明:

$x_1, x_2, \dots, x_k \in C$, 且 $\theta_k \geq 0$, 且 $\theta_k \geq 0$

$$\theta_1 + \theta_2 + \dots + \theta_k = 1$$

使用迭代的办法, 首先考虑 x_1, x_2 .

$$\theta_1 x_1 + \theta_2 x_2 \quad \theta_1 + \theta_2 = 1 \quad \therefore \theta_1 = 1 - \theta_2$$

$$\Rightarrow \theta_1 x_1 + (1 - \theta_1) x_2 \in C$$

设 $N = \theta_1 x_1 + \theta_2 x_2 \Rightarrow N + \theta_3 x_3$ 为凸组合.

$$\therefore \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 \in C$$

递推可得 $\theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k \in C$.

3. 证明:

设 $z_1, z_2 \in S$, 且 $\begin{cases} z_1 = x_1 + y_1 & x_1 \in S_1, y_1 \in S_2 \\ z_2 = x_2 + y_2 & x_2 \in S_1, y_2 \in S_2 \end{cases}$

$$\begin{aligned} \Rightarrow \lambda z_1 + (1-\lambda)z_2 &= \lambda(x_1 + y_1) + (1-\lambda)(x_2 + y_2) \\ &= [\lambda x_1 + (1-\lambda)x_2] + [\lambda y_1 + (1-\lambda)y_2] \end{aligned}$$

$$x_1, x_2 \in S_1; y_1, y_2 \in S_2$$

$$\Rightarrow \lambda x_1 + (1-\lambda)x_2 \in S_1, \lambda y_1 + (1-\lambda)y_2 \in S_2$$

$$\Rightarrow [\lambda x_1 + (1-\lambda)x_2] + [\lambda y_1 + (1-\lambda)y_2] \in S$$

$$\therefore \lambda z_1 + (1-\lambda)z_2 \in S \quad \Rightarrow S \text{ 为凸集.}$$

4. 证明:

$$\begin{aligned} \|\theta x_1 + (1-\theta)x_2 - x_0\|_2 &\leq \|\theta(x_1 - x_0)\|_2 + \|(1-\theta)(x_2 - x_0)\|_2 \\ \Rightarrow \|\theta(x_1 - x_0)\|_2 + \|(1-\theta)(x_2 - x_0)\|_2 &= \theta\|x_1 - x_0\|_2 + (1-\theta)\|x_2 - x_0\|_2 \end{aligned}$$

$$\because x_1, x_2 \in B(x_0, r) \Rightarrow \|x_1 - x_0\|_2 \leq r \text{ 且 } \|x_2 - x_0\|_2 \leq r$$

$$\Rightarrow \theta\|x_1 - x_0\|_2 + (1-\theta)\|x_2 - x_0\|_2 \leq \theta r + (1-\theta)r = r$$

$$\therefore \|\theta x_1 + (1-\theta)x_2 - x_0\|_2 \leq r$$

$$\text{因此 } \theta x_1 + (1-\theta)x_2 \in B(x_0, r)$$

即 $B(x_0, r)$ 为凸集.